

Influence of Ball Reception on Visual Tracking Performance of Soccer Players in a Virtual Reality Multiple-Player-Tracking Task

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Figure 1: The visual tracking task during soccer backward passing situations simulated in a CAVE.

ABSTRACT

The multitasking aspect of soccer is rarely considered to study the relationship between expertise and visual attention. This study examined the ability of 24 soccer players to visually track teammates and opponents in simulated game situations within a CAVE, with or without the requirement to intercept a virtual ball. No significant decrease in visual tracking performance was observed when an actual motor interception was required. Additionally, no significant differences were found between participants competing at the national level and those at lower levels. Further research is needed to better understand how visual attention contributes to expertise in soccer.

Index terms: Soccer, Visual attention, Virtual Reality, CAVE.

1 INTRODUCTION

Soccer players must be aware of the positions of their teammates and opponents to make appropriate decisions while running and/or interacting with the ball. Directly assessing the ability to allocate visual attention between these activities in real games is challenging. Standardized and reproducible off-field exercises are necessary to objectively measure differences in perceptual-cognitive skills between players, and track their development over time. In this context, the Multiple-Object-Tracking (MOT) paradigm is commonly used to investigate the relationship between team sports expertise and dynamic visual attention [1]. MOT can be described as a visual tracking task requiring the monitoring of several visual cues for seconds in a bounded space shared by distracting objects of similar appearance. It is used to study expertise because the induced attentional demands, and thus the involved perceptual-cognitive mechanisms, are assumed to overlap

those required to perceive multiple elements in complex and dynamic situations of everyday life.

However, concerns are raised regarding the ecological validity of traditional visual tracking tasks in sports performance research, i.e., with participants in a fixed position to monitor randomly moving objects on a frontal plane. If task constraints do not correspond to the field constraints under which perceptual-cognitive mechanisms operate, the observed intra- or inter-individual differences may be underestimated. To address this mismatch, a Multiple-Player-Tracking (MPT) version has been proposed, embedding the tracking task within a virtual field to face simulations of soccer game scenarios [2]. Using a Virtual Reality (VR) headset with an adaptive point of view, participants could naturally move their head and body to visually explore their surroundings. Nevertheless, participants in this context only had to track virtual teammates and opponents. Soccer players had to perceive their surrounding while moving or interacting with the ball in real game settings. This multitasking aspect of soccer is rarely considered in visual tracking tasks when investigating soccer expertise. The dual-task methodology, used to operationalize multitasking, could represent an additional step toward improving the representativeness of evaluation constraints and enhancing the discrimination of expertise [3].

The current study adapts the MPT task to simulate the constraints of backward passing situations in a CAVE. From the perspective of the ball receiver, participants engage in two tasks inherent to soccer: monitoring teammates and opponents while intercepting a virtual ball (Figure 1). The objectives were to examine (i) the interception effect, (ii) the expertise effect and (iii) the interaction of both effects on (a) visual tracking performance and (b) global displacement in soccer players.

2 METHODS

2.1 Participants

Twelve players competing in national leagues, High-Skilled group (HS), and 12 others in regional or district leagues, Less-Skilled group (LS) were included in this study (26±8 years old). The experimental protocol was validated beforehand by a French national ethics committee in sports sciences (CERSTAPS).

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2.2 Materials

A 12m x 4m x 4m CAVE was used to display the immersive virtual environment. Participants were equipped with stereoscopic VR goggles, a hand controller, and passive clusters of markers on the ankles. The virtual environment, generated by Unity engine, represented soccer stadium with 21 virtual players modelled as simplified mannequins, opponents in blue and teammates in red (Figure 1). The ball and players were animated according to the trajectories extracted from 12 backward passing situations recorded during professional games.

2.3 Procedure

The participants' role was that of the ball receiver in a backward position of the field, such as a goalkeeper or a central defender. They had to start each trial from an initial position (the gray circle in figure 1), but were then free to move around the CAVE's floor. The 12 situations of 10 seconds were repeated in 4 experimental conditions in random order. Depending of the condition, participants were required to complete one or both of the following tasks. A visual tracking task: Three given target-players had to be visually monitored for a complete trial (10s). An interception task: The virtual ball was passed by a teammate to the left or right of the participant's initial position, from 0.8 to 1.2 m, and it had to be intercepted using one of the feet.

In a dual-task condition (DT-TrackPass), participants were required to complete both tasks simultaneously (see Video 1 in supplementary materials). In a single tracking condition (ST-Track), participants exclusively engaged in the visual tracking task without the presence of the virtual ball. In a single interception task (ST-Pass) condition, participants were required solely to intercept the virtual ball trajectory. In a tracking-no-pass condition (TrackNoPass), the virtual ball remained in the environment, but no pass occurred in the direction of the participant during the visual tracking task.

The visual tracking performance was measured as the ratio of successfully tracked target-players. The distance covered during a trial was measured as the cumulative displacement of the head on the horizontal plane.

3 RESULTS

A mixed-model ANOVA revealed a significant main effect of experimental condition, $F(2,44)=30.64$, $p<.001$, but no significant main effect of group, $F(1,22)=0.39$, $p=.540$, and no significant interaction effect, $F(2,44)=0.96$, $p=.392$, on visual tracking performance. Figure 2 shows the significance levels of the pairwise comparisons.

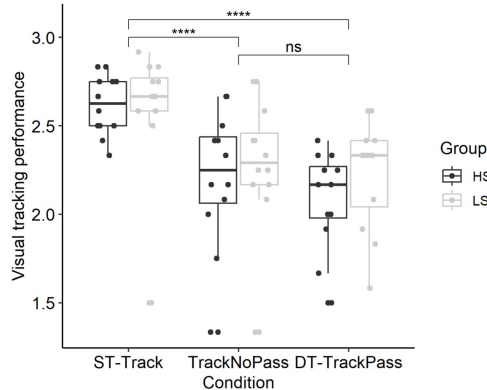


Figure 2: Visual tracking performance by group in the three experimental conditions (****: p -value $<.0001$).

A mixed-model ANOVA revealed a significant main effect of experimental condition, $F(3,66)=66.91$, $p<.001$, but no significant main effect of group, $F(1,22)=0.003$, $p=.958$, and no significant interaction effect, $F(3,66)=0.06$, $p=.981$, on covered distance. Significance levels of pairwise statistics are illustrated in figure 3.

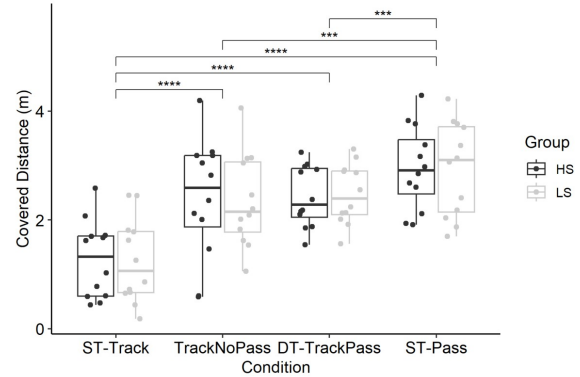


Figure 3: Covered distance by group in the four experimental conditions (***: p -value $<.001$ and ****: p -value $<.0001$).

4 CONCLUSION

Tracking the ball in addition to the virtual players reduced participants' tracking performance. While this could simply be attributed to the additional visual cue to monitor, the longer covered distance suggests that the presence of the ball affects participants' behaviour, which may, in turn, impact visual tracking performance. Although self-motion might negatively influence tracking performance [4], participants naturally engaged in movement to remain ready to interact with the ball.

Contrary to expectations, the imposed constraints did not allow high-skill soccer players to outperform their lower-skill counterparts. Furthermore, no significant differences in visual tracking performance or distance covered were observed when motor execution for ball interception was required. As ball interceptions were almost always successful across trials (98%), the interception task might not have been challenging enough to compete with the visual tracking task. While task complexity may moderate the expertise effect in visual tracking tasks [1], this limitation should be addressed in future research to better explore the relationship between visual attention and expertise.

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